A M **Aahad**

Computer Vision · Machine Learning

3405 NW Orchard Ave Apt 153, Corvallis, OR 97330

+1 (541) 908-6948 | aahada@oregonstate.edu | viper4717 | A M Aahad

Summary

Third-year PhD student in Computer Vision and Machine Learning, currently an AI Engineering Intern at HP Inc., working on multi-camera 3D geometry for conferencing systems. Former software engineer at Samsung, delivering flagship applications for millions of users. Skilled at translating advanced ML research into scalable, real-world solutions across robotics, vision, and edge AI systems.

Education

Oregon State University

Corvallis, Oregon, USA

Ph.D. in Computer Science — *September 2024 - Present*

- **Advisor:** Prof. Sinisa Todorovic
- **CGPA: 3.97**/4.00

Oregon State University

Corvallis, Oregon, USA

M.S. in Computer Science — *September 2024 - June 2026*

- **Advisor:** Prof. Sinisa Todorovic
- **CGPA: 3.97**/4.00
- **Selected Courses:** Machine Learning, Deep Learning, Computer Vision I, Computer Vision II

University of Dhaka

Dhaka, Bangladesh

B.S. in Computer Science and Engineering — *January 2018 - January 2023*

- **Advisor:** Prof. Chowdhury Farhan Ahmed, Dr. Md Samiullah
- **CGPA: 3.75**/4.00
- **Rank: 2**/54

Publications

- **A M Aahad**, Khondker Bin Yamin, Md Samiullah, Chowdhury Farhan Ahmed, Carson K Leung, Evan WR Madill, Adam GM Pazdor. Connectable and Independent Junction Tree-based Compilation Technique of Object-oriented Bayesian Networks. ICTAI'2023
- Inzamamul Alam, Md Samiullah, SM Asaduzzaman, Upama Kabir, **A M Aahad**, Simon S Woo. MIRACLE: Malware Image Recognition and Classification by Layered Extraction. Data Mining and Knowledge Discovery

Professional Experiences

HP Inc.

Palo Alto, California, USA

Artificial Intelligence Engineering Intern — *June 2026 - Present*

- Improved surface-detection accuracy by building a depth-based plane-fitting pipeline for real-time perspective correction.
- Enhanced rectification precision with an edge- and vanishing-point-based orientation correction algorithm.
- Advanced content-capture automation by integrating open-vocabulary object detection for region-of-interest localization.
- Enabled freeform surface capture by adding a manual drag-to-select tool for rectifying any user-chosen region on a plane.

Oregon State University

Corvallis, Oregon, USA

Graduate Research Assistant — *September 2025 - March 2026*

- Conducting applied research in Computer Vision for agricultural automation under an NSF-funded project.
- Improved robotic harvester accuracy through segmentation and depth-based mass flow estimation models for blueberries on conveyors.
- Enhanced selective pruning performance in robotic pruning systems with an instance segmentation framework for tree branch detection.
- Advancing scalable AI solutions for precision agriculture using deep models (Mask2Former, SAM2, YOLO, SegFormer).

Samsung R&D Institute Bangladesh

Dhaka, Bangladesh

Full-time Software Engineer — *February 2023 - August 2024*

- Core engineer (in a 3-member team) for Samsung's flagship "Galaxy Buds" Windows application, used globally by millions of users.
- Led feature development and integration for Buds3/Buds3 Pro and Buds FE, contributing to a successful global product launch.
- Developed the search module and implemented OneUI 6, Quick Controls, and LE Audio features, ensuring seamless compatibility.
- Resolved 90+ PLM issues, refactored core modules and eliminated redundancies, significantly improving app stability and user satisfaction.
- Leveraged UWP, Windows SDK, and WPF with C# / XAML to deliver production-ready UI and device communication layers.

Academic Experiences

Controlling Structured Explanations via Shapley Values

Corvallis, Oregon, USA

Advised by Prof. Sinisa Todorovic — *September 2024 - May 2025*

- Proposed a Shapley value-based framework to analyze and regulate structured explanations of deep networks.
- Formulated an interaction-regularized training objective that controls explanation scale, improving interpretability during learning.
- Validated strong correlations between interaction strength and explanation fidelity across CNNs and Transformers.

MIRACLE: Malware Image Recognition and Classification by Layered Extraction

Dhaka, Bangladesh

Advised by Dr. Md Samiullah — *August 2023 - December 2023*

- Published in Data Mining and Knowledge Discovery (2024).
- Enabled fast and accurate identification of malware families through a lightweight CNN framework for malware image classification.
- Integrated L2-norm CycleGAN augmentation to balance imbalanced datasets, preserving structural integrity and improving model robustness.
- Outperformed SOTA methods such as InceptionV3, ResNet50, and EfficientNet on MalImg and Microsoft-Big with 99.8% accuracy.

Efficient Compilation Algorithm for Object-oriented Bayesian Networks

Dhaka, Bangladesh

Advised by Prof. Chowdhury Farhan Ahmed, Dr. Md Samiullah — *September 2022 - April 2023*

- Published in International Conference on Tools for Artificial Intelligence (ICTAI'2023).
- Proposed a Connectable and Independent Junction Tree (CIC) algorithm to compile OOBNs without flattening.
- Enabled reuse of component-level junction trees, reducing compilation time and improving scalability for large networks.
- Achieved exponential reduction in compilation complexity compared to conventional BN flattening methods.

Teaching Experiences

Oregon State University

Corvallis, Oregon, USA

Graduate Teaching Assistant — CS 271 (Computer Arch. & Assembly Language) & CS 362 (Software Engineering II) — *Sep 24 - Jun 25, Apr 26 - Jun 26*

- Held live weekly office hours, assisting up to 10 students simultaneously with questions on course material and homework.
- Graded weekly assignments, midterms, and finals for 250+ submissions per course, providing detailed feedback within 12 hours.
- Enhanced homework clarity and course materials through actionable feedback, improving student comprehension and learning outcomes.
- Ensured smooth operations and consistent student progress through punctual and organized course support.

Service

Co-Reviewer — *2025*

- International Conference on Computer Vision (ICCV)
- Neural Information Processing Systems (NIPS)

Skills

Computer Vision	Video Instance Segmentation, Vision–Language Models (VLMs), Structured Visual Reasoning
Artificial Intelligence	Machine Learning, Explainable AI (XAI), Generative AI, Optimization
Programming	Python, PyTorch, C#, C++, Javascript, Java, Unix Environment, Software Engineering
Research	Submission (NIPS, ICLR, ICML), Cross-Group & Cross-Cultural Collaboration
Soft Skills	Adaptability, Punctuality, Patience and Perseverance, Problem Solving Skills